

Integral-Based Event-Triggered Fault Detection Filter Design for Unmanned Surface Vehicles

Xudong Wang, Zhongyang Fei , Huijun Gao , and Jinyong Yu 

Abstract—This paper is concerned with the event-triggered fault detection filter (FDF) design for an unmanned surface vehicle (USV) under the network environment. A framework of fault detection is established for a USV subject to wave-induced disturbance and actuator failures, in which an FDF is utilized to construct a residual model and an integral-based event generator is introduced to save communication resources. Compared with the traditional instantaneous value based event-triggering scheme and periodic sampling, the proposed event-triggering mechanism can not only reduce bandwidth utilization of the network more significantly, but also get rid of the Zeno phenomenon fundamentally. The event-triggering scheme and the FDF are co-designed. Finally, the efficient performance of the proposed fault detection method based on integral-based event-triggering scheme is illustrated by simulation.

Index Terms—Fault detection, integral-based event trigger, unmanned surface vehicle (USV).

I. INTRODUCTION

TAKING advantages of satellite-based data communications, wide area communication technologies, and high-accuracy positioning systems, unmanned surface vehicles (USVs) have been developed rapidly in the past few years, since their small size, good concealment, low cost, and powerful field operational advantages [1]–[3]. Some researchers have studied the possibility of using USVs within a wide range of roles, including hydrographic survey, environmental monitoring and sampling, pollutant tracking, coastal surveillance, and rescuing [4]–[9].

It needs to be emphasized that since wave erosion, seawater erosion, extremely low operating temperature, long work hours, etc., the USV system will be unavoidably subjected to failures, such as saturation, stuck, loss effectiveness of actuator, etc. Thus, in order to improve the reliability of system, it is essential to detect the occurrence of system faults as they occur [10]. The fault detection problem has been paid much attention

by researchers and fruitful results can be found in the literature over the last decades. Li and Yang [11] studied the fault detection for Takagi–Sugeno fuzzy systems with frequency range limitation. Cai *et al.* [12] introduced a novel model-based fault detection approach for a class of linear distributed parameter systems. Castillo and Zufiria [13] proposed a novel fault detection approach for continuous-time stochastic dynamic systems. Zhang and Yang [14] proposed a fault detection method based on the interval observers for a class of discrete-time systems.

It is worth mentioning that USVs are working under network environment and data exchange are realized by a wireless communication network. That is, system components including sensor, controller/filter, actuator, etc., are connected through communication network, meanwhile, the controller/filter is usually located in a remote mother ship-based or land-based station [15]. Thus, it is meaningful to consider the network-induced phenomenon while investigating the design method of network-based fault detection filter (FDF). In order to deal with the limited bandwidth of the network, event-triggering scheme, as an attractive alternative to time-driven mechanism such as periodic sampling, has attracted much attention from researchers in recent years. Zhang *et al.* [16] addressed the distributed H_∞ filter design for a cloud-aided active semivehicle suspension system under event-triggering scheme. Chen *et al.* [17] studied the H_∞ load frequency control for networked power systems by adaptive event-triggering sampling. Li and Shi [18] investigated the event-triggered robust model predictive control for nonlinear systems subject to bounded disturbance. However, only a few results on event-triggered fault detection have been published. Liu *et al.* [19] studied the fault detection observer design under the event-triggering scheme for a class of nonlinear networked systems subject to output saturation. Ning *et al.* [20] addressed the co-design of an FDF and controller for uncertain discrete-time stochastic systems by event-triggering mechanism. Wang *et al.* [21] investigated the event-triggered FDF and controller co-design for a continuous-time system subject to biased sensor faults where the packet dropouts and the network-induced delays were considered.

However, event-triggering schemes in the aforementioned works are all depends on the measured outputs directly. However, the existence of noise and disturbance in operating environment of the USV system and the interference of measurements may cause unnecessary data transmissions under the traditional event-triggering scheme. To tackle this problem, Mousavi and Marquez [22] proposed an integral-based event-triggering mechanism via convex optimization. Instead

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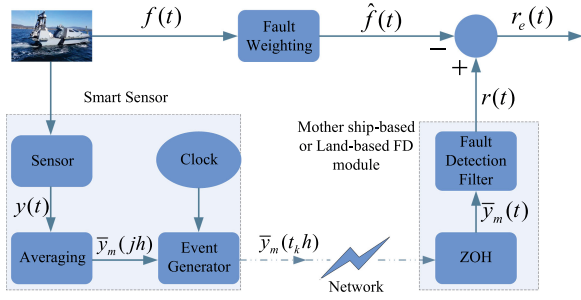


Fig. 1. Frame of fault detection for a USV with the event-triggering scheme.

of measured output, the average of measurements is used to test the event-triggered condition (ETC) and transmitted through communication network. Motivated by this, we would like to investigate the fault detection for a USV system under integral-based event-triggering mechanism to both reduce the data package transmissions and the impact of noise and disturbance.

The contributions of this paper are as follows.

- 1) A novel framework of fault detection for a USV system is established by introducing integral-based event-triggering mechanism.
- 2) The average of measured outputs over a specific time interval is used as the input of integral-based ETC and FDF, which has potential ability to reduce the data package transmission and to attenuate the impact of noise and disturbance.
- 3) The criterion is proposed for the co-design of ETC and FDF.

The rest of this paper is organized as follows. Section II presents the framework of fault detection under integral-based event-triggering scheme. The coordinated design of ETC and FDF is presented in Section III. In Section IV, simulation results are provided to show the effectiveness of the proposed method. Conclusions are made in Section V.

Notations: The following notations are used throughout this paper. For a matrix M , its inverse and transpose are denoted by M^{-1} and M^T , respectively; the notation $M > 0$ means that matrix M is positive definite and vice versa; $[X]_s$ is the simplified notation of $X + X^T$; $\{\cdot\}$ denotes the entries of a matrix implied by symmetry; $\text{diag}\{\cdot\}$ stands for a block-diagonal matrix; and $\mathcal{L}_2[0, \infty)$ denotes the linear space of square integrable vectors over the time interval $[0, \infty)$. The space of functions $\phi: [a, b] \rightarrow \mathbb{R}^n$, which are absolutely continuous on $[a, b]$, have a finite $\lim_{\theta \rightarrow b^-} \phi(\theta)$ and have square integrable first-order derivatives are denoted by $W[a, b]$ with the norm $\|\phi\|_W = \max_{\theta \in [a, b]} |\phi(\theta)| + [\int_a^b |\dot{\phi}(s)|^2 ds]^{\frac{1}{2}}$. All matrices in this paper are supposed to be compatible for algebraic operation if their dimensions are not explicitly stated.

II. SYSTEM MODELING AND PROBLEM FORMULATION

The event-based fault detection framework is depicted in Fig. 1 for a networked USV system in the continuous time domain. The USV and the remote FDF are connected through a wireless communication network. To save limited resource,

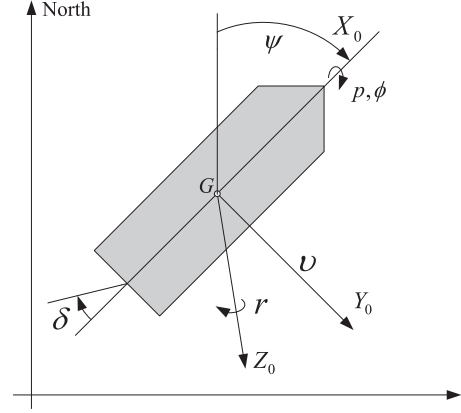


Fig. 2. Motion coordinate system of a USV.

the integral-based event-triggering scheme is introduced, such that the average of measurements is transmitted only when the predefined ETC is violated.

A. System Modeling

The motion of a USV in 6 DOF includes sway, yaw, roll, heave, surge, and pitch [23]. For simplicity, only the motion in sway, yaw, and roll is considered in this paper, whereas the influence of heave, surge, and pitch is modeled as external disturbance. The motion coordinate is displayed in Fig. 2, and the corresponding motion equations are described as follows [15]:

$$\begin{cases} \dot{v}(t) = -\frac{1}{T_v}v(t) + \frac{K_{dv}}{T_v}\delta(t) \\ \dot{r}(t) = \frac{K_{vr}}{T_r}v(t) - \frac{1}{T_r}r(t) + \frac{K_{dr}}{T_r}\delta(t) + \frac{1}{T_r}\omega_\psi(t) \\ \dot{p}(t) = \omega_n^2 K_{vp}v(t) - 2\zeta\omega_n p(t) - \omega^2\phi(t) \\ \quad + \omega_n^2 K_{dp}\delta(t) + \omega_n^2\omega_\phi(t) \\ \dot{\phi}(t) = p(t) \\ \dot{\psi}(t) = r(t) \end{cases} \quad (1)$$

where $v(t)$, $r(t)$, $\psi(t)$, $p(t)$, $\phi(t)$, and $\delta(t)$ denote the sway velocity caused by the rudder motion, yaw velocity, heading angle, roll velocity, roll angle, rudder angle, respectively; $\omega_\psi(t)$ and $\omega_\phi(t)$ denote the wave-induced disturbance on $\psi(t)$ and $\phi(t)$, respectively; K_{vr} , K_{vp} , K_{dv} , K_{dr} , and K_{dp} denote the given gains; T_v and T_r denote the time constants; and ζ and ω_n denote the damping coefficient and the undamped natural frequency, respectively.

First, define the system state vector as $x(t) = \text{col}\{v(t), r(t), \psi(t), p(t), \phi(t)\}$ with $x(t) \in \mathbb{R}^n$, the disturbance vector as $\omega(t) = \text{col}\{\omega_\psi(t), \omega_\phi(t)\}$ with $\omega(t) \in \mathbb{R}^d$, and the system input as $u(t) = \delta(t)$ with $u(t) \in \mathbb{R}^m$. Then, the dynamic equation (1) can be rewritten as

$$\dot{x}(t) = A_0x(t) + B_0u(t) + B_1\omega(t) \quad (2)$$

where

$$A_0 = \begin{bmatrix} -\frac{1}{T_v} & 0 & 0 & 0 & 0 \\ \frac{K_{vr}}{T_r} & -\frac{1}{T_r} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \omega_n^2 K_{vp} & 0 & 0 & -2\zeta\omega_n & -\omega_n^2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$B_0 = \begin{bmatrix} \frac{K_{dv}}{T_v} & \frac{K_{dr}}{T_r} & 0 & \omega_n^2 K_{dp} & 0 \end{bmatrix}^T$$

$$B_1 = \begin{bmatrix} 0 & \frac{1}{T_r} & 0 & 0 & 0 \\ 0 & 0 & 0 & \omega_n^2 & 0 \end{bmatrix}^T.$$

In this paper, we assume that the system has been stabilized by a controller $u(t) = Kx(t)$. To obtain feedback controller, a simple way to choose matrix K is to make the matrix $A_0 + B_0K$ Hurwitz. Meanwhile, consider the actuator faults caused by the component torn-and-worn factors, etc., the system (2) can be rewritten as

$$\begin{aligned} \dot{x}(t) &= Ax(t) + B_1\omega(t) + B_2f(t) \\ y(t) &= Cx(t) \end{aligned} \quad (3)$$

where $y(t) \in \mathbb{R}^p$ is the measurement output vector, $f(t) \in \mathbb{R}^l$ denotes a possible fault, and $\omega(t)$, $f(t)$ belong to $\mathcal{L}_2[0, \infty)$. The closed-loop system matrix $A = A_0 + B_0K$. The matrices B_2 and C are known constant matrices with appropriate dimensions.

B. Fault Weighting System

In order to improve the fault detection performance, a weighting fault function is usually included in the framework of fault detection, which can be represented as $\hat{f}(z) = W(z)f(z)$, where $W(z)$ is a weighting matrix reflecting the frequency weighting of the fault signal [24]. One of the state-space realizations of $W(z)$ is shown as follows:

$$\begin{aligned} \dot{x}_w(t) &= A_w x_w(t) + B_w f(t) \\ \hat{f}(t) &= C_w x_w(t) + D_w f(t) \end{aligned} \quad (4)$$

where $x_w(t) \in \mathbb{R}^w$ is the state vector of fault weighting system; $\hat{f}(t) \in \mathbb{R}^l$ is the weighting fault signal; and A_w , B_w , C_w , and D_w are predefined constant matrices.

C. Fault Detection Filter

The main objective of this paper is to design an integral-based event-triggered fault detection scheme. An auxiliary dynamic system called residual generator is constructed to generate the residual signal that is used to decide whether or not system faults have occurred. The following FDF is constructed to generate the residual signal:

$$\begin{aligned} \dot{\hat{x}}(t) &= A_f \hat{x}(t) + B_f \bar{y}_m(t) \\ r(t) &= C_f \hat{x}(t) + D_f \bar{y}_m(t) \end{aligned} \quad (5)$$

where $\hat{x}(t) \in \mathbb{R}^{n+w}$ is the filter state vector, $r(t) \in \mathbb{R}^l$ is the residual signal, and $\bar{y}_m(t) \in \mathbb{R}^p$ denotes the input of the FDF (which is influenced by event-triggering scheme). The matrices A_f , B_f , C_f , and D_f are filter matrices with appropriate dimensions to be designed later.

D. Event Generator

The main purpose of introducing event-triggering scheme in fault detection framework is to save the limited network communication resources. In this paper, we suppose that the event generator module is clock driven, i.e., examines the ETC periodically with the period h . Instead of instant measurements, the average of the measurements over the time interval T is used as the input of event generator module and transmitted through the wireless communication network, where T is the length of integral interval. Define $\bar{y}_m(jh)$ as the input of event generator, with

$$\bar{y}_m(jh) = \frac{1}{T} \int_{jh-T}^{jh} y(s) ds, \quad j \in \mathbb{N}. \quad (6)$$

The set of transmission instants is presented by $\{t_k h | t_k \in \mathbb{N}\}$, where t_k is decided by the following periodic integral-based ETC (PIB-ETC):

$$t_{k+1} = t_k + \min_{l>0} \{l | \Delta_{\bar{y}_m}^T \Omega \Delta_{\bar{y}_m} > \varepsilon \bar{y}_m(ih)^T \Omega \bar{y}_m(ih)\} \quad (7)$$

with $\Delta_{\bar{y}_m} = \bar{y}_m(ih) - \bar{y}_m(t_k h)$ and $ih = t_k h + lh$.

Note that the ETC in (7) ensures that the interevent time is lower bounded by $h > 0$, which means that the Zeno phenomenon can be eliminated technically.

Remark 1: Compared with the traditional periodic ETC (PETC) in [25] and [26], which can be presented as

$$t_{k+1} = t_k + \min_{l>0} \{l | \Delta_y^T \Omega \Delta_y > \varepsilon y(ih)^T \Omega y(ih)\} \quad (8)$$

with $\Delta_y = y(ih) - y(t_k h)$ and $ih = t_k h + lh$, the proposed integral-based ETC has potential ability to reduce unnecessary data transmissions caused by disturbance and noise due to the introduction of $\bar{y}_m(jh)$. The average of outputs over the time interval T is used as the input of the event generator. This could avoid the unnecessary triggering events caused by disturbance and noise significantly, especially when the measured outputs are contaminated by measurement noise, which is usually assumed to be zero mean white Gaussian noise. More details will be illustrated in Section IV.

Taking the network-induced delay and the behavior of the zero-order-holder (ZOH) into account, the input of FDF $\bar{y}_m(t)$ can be presented by

$$\bar{y}_m(t) = \bar{y}_m(t_k h), \quad t \in [t_k h + \eta_k, t_{k+1} h + \eta_{k+1})$$

where $\eta_k > 0$ is the transmission delay and $\eta_m \leq \eta_k \leq \eta_M$. Define the following subintervals on $[t_k h + \eta_k, t_{k+1} h + \eta_{k+1})$:

$$\begin{cases} \Upsilon_0 &= [t_k h + \eta_k, t_k h + h + \eta_k) \\ \Upsilon_1 &= [t_k h + h + \eta_k, t_k h + 2h + \eta_k) \\ &\vdots \\ \Upsilon_{j_M^*} &= [t_k h + j_M^* h + \eta_k, t_{k+1} h + \eta_{k+1}) \end{cases} \quad (9)$$

with $j_M^* = \min\{j|t_k h + (j+1)h + \eta_k \geq t_{k+1}h + \eta_{k+1}\}$. The piecewise time-varying delay $\tau(t)$ and the error vector $e(t)$ are defined as follows:

$$\tau(t) = \begin{cases} t - t_k h, & t \in \Upsilon_0 \\ t - t_k h - h, & t \in \Upsilon_1 \\ \vdots \\ t - t_k h - j_M^* h, & t \in \Upsilon_{j_M^*} \end{cases} \quad (10)$$

$$e(t) = \begin{cases} \bar{y}_m(t_k h) - \bar{y}_m(t_k h), & t \in \Upsilon_0 \\ \bar{y}_m(t_k h + h) - \bar{y}_m(t_k h), & t \in \Upsilon_1 \\ \vdots \\ \bar{y}_m(t_k h + j_M^* h) - \bar{y}_m(t_k h), & t \in \Upsilon_{j_M^*} \end{cases} \quad (11)$$

It should be mentioned that $\tau(t)$ is not the network-induced delay but a time-varying artificial delay. Then, the input of the FDF $\bar{y}_m(t)$ can be rewritten as

$$\bar{y}_m(t) = \bar{y}_m(t_k h) = \bar{y}_m(t - \tau(t)) - e(t). \quad (12)$$

Denote $\tau_m = \eta_m$, $\tau_M = h + \eta_M$, and $d = \tau_M - \tau_m$. It follows that $\tau_m \leq \tau(t) \leq \tau_M$.

Before proceeding, an approximation of the integral term $\bar{y}_m(t - \tau(t))$ appeared in (12) is introduced to simplify the process. By Simpson's rule [27], we have

$$\int_{t-\tau(t)-T}^{t-\tau(t)} y(s)ds \approx \frac{T}{6}[y(t - \tau(t)) + 4y(t - \tau(t) - T/2) + y(t - \tau(t) - T)]. \quad (13)$$

The approximate error is given by $\frac{1}{90}(\frac{T}{2})^5|y^4(\xi)|$, where ξ is some point in the time interval $[t - \tau(t) - T, t - \tau(t)]$. Then, the integral term $\bar{y}_m(t - \tau(t))$ is approximated as follows:

$$\bar{y}_m(t - \tau(t)) = C[\alpha_1 x(t - \tau(t)) + \alpha_2 x(t - \tau(t) - T/2) + \alpha_3 x(t - \tau(t) - T)] \quad (14)$$

where $\alpha_1 = 1/6$, $\alpha_2 = 2/3$, and $\alpha_3 = 1/6$.

Remark 2: Among the many numerical integration techniques, Simpson's rule is adopted in this work as a good tradeoff between approximation precision and computing complexity. It is assumed that the approximation error $\frac{1}{90}(\frac{T}{2})^5|y^4(\xi)|$ is negligible. This assumption is reasonable when the provided integral period T is small enough and the signal $y(t)$ is bounded.

E. Residual Evaluation

In order to detect the system faults as they occur, the residual evaluation function $\chi(t)$ and the threshold χ_{th} for $\chi(t)$ are constructed such that an alarm will be generated when $\chi(t)$ is larger than χ_{th} . Here, we consider the evaluation function as follows:

$$\chi(t) = \sqrt{\frac{1}{t} \int_0^t r^T(s)r(s)ds}. \quad (15)$$

The residual evaluation function threshold χ_{th} is chosen as

$$\chi_{th} = \sup_{\omega(t) \in \mathcal{L}_2, f(t)=0} \chi(t). \quad (16)$$

Then, the fault detection logic can be taken as

$$\begin{cases} \chi(t) > \chi_{th} \Rightarrow \text{with faults} \Rightarrow \text{alarm} \\ \chi(t) \leq \chi_{th} \Rightarrow \text{no faults} \end{cases} \quad (17)$$

F. Residual System

From (3)–(18) and (12)–(14), the residual system can be obtained as

$$\begin{aligned} \dot{\xi}(t) &= A_1 \xi(t) + A_2 \xi(t - \tau(t)) + A_3 \xi(t - \tau(t) - T/2) \\ &\quad + A_4 \xi(t - \tau(t) - T) + B \vartheta(t) + E_1 e(t) \\ r_e(t) &= C_1 \xi(t) + C_2 \xi(t - \tau(t)) + C_3 \xi(t - \tau(t) - T/2) \\ &\quad + C_4 \xi(t - \tau(t) - T) + D \vartheta(t) + E_2 e(t) \end{aligned} \quad (18)$$

with $\xi(t) = [x_w^T(t) \ x^T(t) \ \hat{x}^T(t)]^T$, $\vartheta(t) = [\omega^T(t) \ f^T(t)]^T$, $r_e(t) = r(t) - \hat{f}(t)$, and

$$\begin{aligned} A_1 &= \begin{bmatrix} \bar{A}_1 & 0 \\ 0 & A_f \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 0 \\ \alpha_1 \mathcal{A} & 0 \end{bmatrix}, A_3 = \begin{bmatrix} 0 & 0 \\ \alpha_2 \mathcal{A} & 0 \end{bmatrix} \\ A_4 &= \begin{bmatrix} 0 & 0 \\ \alpha_3 \mathcal{A} & 0 \end{bmatrix}, \bar{A}_1 = \begin{bmatrix} A_w & 0 \\ 0 & A \end{bmatrix}, \bar{B} = \begin{bmatrix} 0 & B_w \\ B_1 & B_2 \end{bmatrix} \\ B &= \begin{bmatrix} \bar{B} \\ 0 \end{bmatrix}, E_1 = \begin{bmatrix} 0 \\ -B_f \end{bmatrix}, \mathcal{A} = [0 \quad B_f C] \\ C &= [0 \quad D_f C], C_1 = [\bar{C}_1 \quad C_f], C_2 = [\alpha_1 \mathcal{C} \quad 0] \\ C_3 &= [\alpha_2 \mathcal{C} \quad 0], C_4 = [\alpha_3 \mathcal{C} \quad 0], \bar{C}_1 = [-C_w \quad 0] \\ D &= [0 \quad -D_w], E_2 = -D_f. \end{aligned}$$

The main purpose of this paper is to tackle with the fault detection problem under integral-based event-triggering scheme (7), i.e., design FDF in form of (5) and ETC in form of (7), simultaneously, such that the following holds.

- 1) The residual system (18) is asymptotically stable.
- 2) The residual system (18) has an H_∞ performance level γ , under zero initial conditions, i.e., for all nonzero $\vartheta(t) \in \mathcal{L}_2$, the following inequality is satisfied:

$$\int_0^\infty r_e^T(t)r_e(t)dt \leq \gamma^2 \int_0^\infty \vartheta^T(t)\vartheta(t)dt. \quad (19)$$

Before proceeding further, we recall the following lemmas, which play important roles in the derivation of our results.

Lemma 1 (see [28]): For a positive definite matrix $R > 0$, and an integrable function $\{w(u)|u \in [a, b]\}$, the following

inequality holds:

$$\begin{aligned} & - \int_a^b w^T(\alpha) R w(\alpha) d\alpha \\ & \leq \frac{1}{a-b} \left(\int_a^b w(\alpha) d\alpha \right)^T R \left(\int_a^b w(\alpha) d\alpha \right). \end{aligned} \quad (20)$$

Lemma 2 (see [29]): For any matrix $M = \begin{bmatrix} R & G \\ * & R \end{bmatrix} \geq 0$, $R > 0$, scalars $0 \leq d(t) \leq d$, and a vector function $\dot{x} : [-d, 0] \rightarrow \mathbb{R}^n$ such that the integration in the following inequality is well defined, thereby holding

$$-d \int_{t-d}^t \dot{x}^T(s) R \dot{x}(s) ds \leq \vartheta^T(t) Z \vartheta(t) \quad (21)$$

where

$$\vartheta^T(t) = [x^T(t) \quad x^T(t-d(t)) \quad x^T(t-d)]$$

$$Z = \begin{bmatrix} -R & R-G & G \\ * & [G-R]_s & R-G \\ * & * & -R \end{bmatrix}.$$

III. MAIN RESULTS

In this section, the FDF for a USV system is designed under PIB-ETC (7) to detect the system faults as they occur. The following theorems present the method for coordinated design of PIB-ETC and FDF.

Theorem 1: Give positive scalars $\varepsilon, h, \eta_m, \eta_M, T, \gamma$, FDF matrices A_f, B_f, C_f, D_f , and positive definite matrix Ω . The residual system (18) is asymptotically stable and satisfies (19) for all nonzero $\vartheta(t) \in \mathcal{L}_2$, if there exist matrices $P > 0, Q_j > 0, R_j > 0, S_j > 0, M_j > 0, N_j > 0$, and G_j ($j = 0, 1, 2$) with appropriate dimensions satisfying $\begin{bmatrix} N_j & G_j \\ * & N_j \end{bmatrix} > 0$, such that

$$\begin{bmatrix} \Gamma & \Gamma^R & \Gamma^N & \Gamma^S & \Xi^T \\ * & \mathcal{R} & 0 & 0 & 0 \\ * & * & \mathcal{N} & 0 & 0 \\ * & * & * & \mathcal{S} & 0 \\ * & * & * & * & -I \end{bmatrix} < 0 \quad (22)$$

where

$$\begin{aligned} \Gamma &= \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \Gamma_{13} & 0 & PB & PE_1 \\ * & \Gamma_{22} & \Gamma_{23} & \Gamma_{24} & 0 & 0 \\ * & * & \Gamma_{33} & \Gamma_{34} & 0 & 0 \\ * & * & * & \Gamma_{44} & 0 & 0 \\ * & * & * & * & -\gamma^2 I & 0 \\ * & * & * & * & * & -\Omega \end{bmatrix} \\ \Gamma^R &= [\Gamma_0^R \quad \Gamma_1^R \quad \Gamma_2^R], \quad \Gamma^N = [\Gamma_0^N \quad \Gamma_1^N \quad \Gamma_2^N] \\ \Gamma^S &= [\Gamma_0^S \quad \Gamma_1^S \quad \Gamma_2^S], \quad \mathcal{R} = -\text{diag} \{R_0, R_1, R_2\} \\ \mathcal{N} &= -\text{diag} \{N_0, N_1, N_2\}, \quad \mathcal{S} = -\text{diag} \{S_0, S_1, S_2\} \\ \Xi &= [C_1 0 0 0 \quad C_2 \quad C_3 \quad C_4 \quad 0 0 0 \quad D \quad E_2] \end{aligned}$$

with

$$\begin{aligned} \Gamma_{11} &= [PA_1]_s + \sum_{j=0}^2 Q_j^H - R_j^H - \frac{\pi^2}{4} S_i^H \\ \Gamma_{12} &= [R_0^H \quad R_1^H \quad R_2^H] \\ \Gamma_{13} &= \left[PA_2 + \frac{\pi^2}{4} S_0^H \quad PA_3 + \frac{\pi^2}{4} S_1^H \quad PA_4 + \frac{\pi^2}{4} S_2^H \right] \\ \Gamma_{22} &= \text{diag} \{M_0^H - Q_0^H, M_1^H - Q_1^H, M_2^H - Q_2^H\} \\ &\quad - \text{diag} \{R_0^H + N_0^H, R_1^H + N_1^H, R_2^H + N_2^H\} \\ \Gamma_{23} &= \text{diag} \{N_0^H - G_0^H, N_1^H - G_1^H, N_2^H - G_2^H\} \\ \Gamma_{24} &= \text{diag} \{G_0^H, G_1^H, G_2^H\} \\ \Gamma_{33} &= \text{diag} \{[G_0^H - N_0^H]_s, [G_1^H - N_1^H]_s, [G_2^H - N_2^H]_s\} \\ &\quad - \text{diag} \left\{ \frac{\pi^2}{4} S_0^H, \frac{\pi^2}{4} S_1^H, \frac{\pi^2}{4} S_2^H \right\} + \varepsilon C_\alpha^T \Omega C_\alpha \\ \Gamma_{34} &= \text{diag} \{N_0^H - G_0^H, N_1^H - G_1^H, N_2^H - G_2^H\} \\ \Gamma_{44} &= -\text{diag} \{M_0^H + N_0^H, M_1^H + N_1^H, M_2^H + N_2^H\}, \\ \Gamma_j^R &= (\tau_m + jT/2) A^T H^T R_j, \quad \Gamma_j^N = d A^T H^T N_j, \\ \Gamma_j^S &= (\tau_M + jT/2) A^T H^T S_j, \quad H = [I \quad 0], \\ A &= [A_1 0 0 0 \quad A_2 \quad A_3 \quad A_4 \quad 0 0 0 \quad B \quad E_1], \\ C_\alpha &= [\alpha_1 C_0 \quad \alpha_2 C_0 \quad \alpha_3 C_0], \quad C_0 = [0 \quad C \quad 0], \\ R_j^H &= H^T R_j H, \quad Q_j^H = H^T Q_j H, \quad G_j^H = H^T G_j H, \\ S_j^H &= H^T S_j H, \quad M_j^H = H^T M_j H, \quad N_j^H = H^T N_j H. \end{aligned}$$

Proof: Inspired by Zhang and Han[30], consider the following Lyapunov-Krasovskii functional:

$$V(t) = V_1(t) + V_2(t) + V_3(t) + V_4(t) \quad (23)$$

with

$$\begin{aligned} V_1(t) &= \xi^T(t) P \xi(t) \\ V_2(t) &= \sum_{j=0}^2 \int_{t-\tau_m-jT/2}^t \xi^T(s) Q_j^H \xi(s) ds \\ &\quad + \sum_{j=0}^2 \int_{t-\tau_M-jT/2}^{t-\tau_m-jT/2} \xi^T(s) M_j^H \xi(s) ds \\ V_3(t) &= \sum_{j=0}^2 (\tau_m + jT/2) \int_{t-\tau_m-jT/2}^t \int_{t-\tau_m-jT/2}^\theta \xi^T(s) R_j^H \dot{\xi}(s) ds d\theta \\ &\quad + \sum_{j=0}^2 d \int_{t-\tau_M-jT/2}^{t-\tau_m-jT/2} \int_{t-\tau_m-jT/2}^\theta \xi^T(s) N_j^H \dot{\xi}(s) ds d\theta \end{aligned}$$

and a discontinuous term $V_4(t)$ as follows [31]:

$$V_4(t) = \sum_{j=0}^2 (\tau_M + jT/2)^2 \int_{t_k h + i h - jT/2}^t \xi^T(s) S_j^H \dot{\xi}(s) ds \\ - \frac{\pi^2}{4} \int_{t_k h + i h - jT/2}^t \varpi_j^T(s) S_j^H \varpi_j(s) ds, \quad t \in \Upsilon_i$$

where $\varpi_j(s) = \xi(s) - \xi(t_k h + i h - jT/2)$.

Differentiating functions in (23) with respect to t yields

$$\dot{V}_1(t) = 2\xi^T(t) P \Lambda \eta(t)$$

$$\dot{V}_2(t) = \sum_{j=0}^2 \xi^T(t) Q_j^H \xi(t)$$

$$- \sum_{j=0}^2 \xi^T(t - \tau_m - jT/2) (Q_j^H - M_j^H) \\ \times \xi(t - \tau_m - jT/2) \\ - \sum_{j=0}^2 \xi^T(t - \tau_M - jT/2) M_j^H \xi(t - \tau_M - jT/2)$$

$$\dot{V}_3(t) = \sum_{j=0}^2 (\tau_m + jT/2)^2 \dot{\xi}^T(t) R_j^H \dot{\xi}(t) \\ + \sum_{j=0}^2 d^2 \dot{\xi}^T(t) N_j^H \dot{\xi}(t) \\ - (\tau_m + jT/2) \int_{t - \tau_m - jT/2}^t \xi^T(\theta) R_j^H \dot{\xi}(\theta) d\theta \\ - d \int_{t - \tau_M - jT/2}^{t - \tau_m - jT/2} \xi^T(\theta) N_j^H \dot{\xi}(\theta) d\theta \\ \dot{V}_4(t) = \sum_{j=0}^2 (\tau_M + jT/2)^2 \dot{\xi}^T(t) S_j^H \dot{\xi}(t) - \frac{\pi^2}{4} \varpi_j'(t)^T S_j^H \varpi_j'(t)$$

where

$$\eta(t) = \text{col}\{\xi(t), \xi_m(t), \xi_\tau(t), \xi_M(t), \vartheta(t), e(t)\} \\ \xi_m(t) = \text{col}\{\xi(t - \tau_m), \xi(t - \tau_m - T/2), \xi(t - \tau_m - T)\} \\ \xi_\tau(t) = \text{col}\{\xi(t - \tau(t)), \xi(t - \tau(t) - T/2), \xi(t - \tau(t) - T)\} \\ \xi_M(t) = \text{col}\{\xi(t - \tau_M), \xi(t - \tau_M - T/2), \xi(t - \tau_M - T)\} \\ \varpi_j'(t) = \xi(t) - \xi(t - \tau(t) - jT/2).$$

Based on Lemma 1, it holds that

$$- (\tau_m + jT/2) \int_{t - \tau_m - jT/2}^t \xi^T(\theta) R_j^H \dot{\xi}(\theta) d\theta \\ \leq - [\xi(t) - \xi(t - \tau_m - jT/2)]^T R_j^H [\xi(t) - \xi(t - \tau_m - jT/2)].$$

By Lemma 2 and the relation $\tau_m \leq \tau(t) < \tau_M$, we have

$$-d \int_{t - \tau_M - jT/2}^{t - \tau_m - jT/2} \xi^T(\theta) N_j^H \dot{\xi}(\theta) d\theta \leq \eta_j^T(t) \Pi_j \eta_j(t)$$

with

$$\eta_j(t) = \text{col}\{\xi(t - \tau_m - jT/2), \xi(t - \tau(t) - jT/2) \\ \xi(t - \tau_M - jT/2)\} \\ \Pi_j = \begin{bmatrix} -N_j^H & N_j^H - G_j^H & G_j^H \\ * & [G_j^H - N_j^H]_s & N_j^H - G_j^H \\ * & * & -N_j^H \end{bmatrix}.$$

Note that the following inequality is ensured by PIB-ETC (7) for $t \in [t_k h + \eta_k, t_{k+1} h + \eta_{k+1}]$:

$$\varepsilon \bar{y}_m^T(t - \tau(t)) \Omega \bar{y}_m(t - \tau(t)) - e^T(t) \Omega e(t) \geq 0. \quad (24)$$

In the following, we first prove that the residual system (18) with $\vartheta(t) \equiv 0$ is asymptotically stable. In doing so, define $\tilde{\eta}(t) = \text{col}\{\xi(t), \xi_m(t), \xi_\tau(t), \xi_M(t), e(t)\}$. Then, we have

$$\dot{V}(t) \leq \tilde{\eta}^T(t) \tilde{\Psi} \tilde{\eta}(t) \quad (25)$$

where

$$\tilde{\Psi} = \tilde{I} + \sum_{j=0}^2 \tilde{I}_j^R R_j^{-1} \tilde{I}_j^{RT} + \tilde{I}_j^N N_j^{-1} \tilde{I}_j^{NT} + \tilde{I}_j^S S_j^{-1} \tilde{I}_j^{ST}$$

with

$$\tilde{I} = \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \Gamma_{13} & 0 & P E_1 \\ * & \Gamma_{22} & \Gamma_{23} & \Gamma_{24} & 0 \\ * & * & \Gamma_{33} & \Gamma_{34} & 0 \\ * & * & * & \Gamma_{44} & 0 \\ * & * & * & * & -\Omega \end{bmatrix}$$

$$\tilde{I}_j^R = (\tau_m + jT/2) \tilde{\Lambda}^T H^T R_j$$

$$\tilde{I}_j^N = d \tilde{\Lambda}^T H^T N_j$$

$$\tilde{I}_j^S = (\tau_M + jT/2) \tilde{\Lambda}^T H^T S_j$$

$$\tilde{\Lambda} = [A_1 0 0 0 A_2 A_3 A_4 0 0 0 E_1].$$

By Schur complement, $\tilde{\Psi} < 0$ can be guaranteed by (22), which implies that $\dot{V}(t) < 0$. Thus, the residual system (18) is asymptotically stable.

In order to establish the H_∞ performance (19) of the residual system, the following function is constructed:

$$J(t) = \int_0^t r_e^T(s) r_e(s) - \gamma^2 \vartheta^T(s) \vartheta(s) ds. \quad (26)$$

It follows that

$$J(t) = \int_0^t r_e^T(s) r_e(s) - \gamma^2 \vartheta^T(s) \vartheta(s) + \dot{V}(s) ds - V(t) \\ \leq \int_0^t r_e^T(s) r_e(s) - \gamma^2 \vartheta^T(s) \vartheta(s) + \dot{V}(s) ds$$

$$\begin{aligned}
& + \varepsilon \bar{y}_m^T(s - \tau(s)) \Omega \bar{y}_m(s - \tau(s)) - e^T(s) \Omega e(s) ds \\
& \leq \int_0^t \eta^T(s) \Psi \eta(s) ds
\end{aligned} \quad (27)$$

where

$$\begin{aligned}
\Psi = & \Gamma + \sum_{j=0}^2 \Gamma_j^R R_j^{-1} \Gamma_j^{RT} + \Gamma_j^N N_j^{-1} \Gamma_j^{NT} \\
& + \Gamma_j^S S_j^{-1} \Gamma_j^{ST} + \Xi^T \Xi.
\end{aligned}$$

By Schur complement, we can derive that $\Psi < 0$ is equivalent to inequality in (22), i.e., $J(t) < 0$, which guarantees the H_∞ performance (19) of the residual system (18). ■

Note the fact that the FDF parameter matrices are coupled with unknown matrix variables in Theorem 1. The conditions in (22) cannot be handled directly by existing optimization toolbox. Thus, the following theorem is given to deal with FDF and PIB-ETC coordinated design for the residual system (18).

Theorem 2: For given scalars $\varepsilon, h, \eta_m, \eta_M, T$, and γ , the residual system (18) is asymptotically stable with an H_∞ performance index γ , if there exist matrices $P_1 > 0, Q_j > 0, R_j > 0, S_j > 0, M_j > 0, N_j > 0, G_j (j = 0, 1, 2), \Omega > 0, W > 0$, and $\bar{A}_f, \bar{B}_f, \bar{C}_f, \bar{D}_f$ with appropriate dimensions, such that $P_1 - W > 0$ and $\begin{bmatrix} N_j & G_j \\ * & N_j \end{bmatrix} > 0$, satisfying

$$\begin{bmatrix} \bar{\Gamma} & \bar{\Gamma}^R & \bar{\Gamma}^N & \bar{\Gamma}^S & \bar{\Xi}^T \\ * & \mathcal{R} & 0 & 0 & 0 \\ * & * & \mathcal{N} & 0 & 0 \\ * & * & * & \mathcal{S} & 0 \\ * & * & * & * & -I \end{bmatrix} < 0 \quad (28)$$

where

$$\bar{\Gamma} = \begin{bmatrix} \bar{\Gamma}_{11} & \bar{\Gamma}_{12} & \bar{\Gamma}_{13} & 0 & \bar{\Gamma}_{15} & \bar{\Gamma}_{16} \\ * & \bar{\Gamma}_{22} & \bar{\Gamma}_{23} & \bar{\Gamma}_{24} & 0 & 0 \\ * & * & \bar{\Gamma}_{33} & \bar{\Gamma}_{34} & 0 & 0 \\ * & * & * & \bar{\Gamma}_{44} & 0 & 0 \\ * & * & * & * & -\gamma^2 I & 0 \\ * & * & * & * & * & -\Omega \end{bmatrix}$$

$$\bar{\Xi} = [\bar{C}_1 \ \bar{C}_f \ 0 \ 0 \ 0 \ \alpha_1 \bar{C} \ \alpha_2 \bar{C} \ \alpha_3 \bar{C} \ 0 \ 0 \ 0 \ D \ \bar{E}_2]$$

$$\begin{aligned}
\bar{\Gamma}^R = & [\bar{\Gamma}_0^R \ \bar{\Gamma}_1^R \ \bar{\Gamma}_2^R], \quad \bar{\Gamma}^N = [\bar{\Gamma}_0^N \ \bar{\Gamma}_1^N \ \bar{\Gamma}_2^N], \quad \bar{\Gamma}^S \\
= & [\bar{\Gamma}_0^S \ \bar{\Gamma}_1^S \ \bar{\Gamma}_2^S]
\end{aligned}$$

with

$$\begin{aligned}
\bar{\Gamma}_{11} = & \begin{bmatrix} [P_1 \bar{A}_1]_s + \sum_{j=0}^2 (Q_j - R_j - \frac{\pi^2}{4} S_i) & \bar{A}_f + \bar{A}_1^T W \\ * & \bar{A}_f + \bar{A}_f^T \end{bmatrix} \\
\bar{\Gamma}_{13} = & \begin{bmatrix} \alpha_1 \bar{A} + \frac{\pi^2}{4} S_0 & \alpha_2 \bar{A} + \frac{\pi^2}{4} S_1 & \alpha_3 \bar{A} + \frac{\pi^2}{4} S_2 \\ \alpha_1 \bar{A} & \alpha_2 \bar{A} & \alpha_3 \bar{A} \end{bmatrix}
\end{aligned}$$

$$\bar{\Gamma}_{12} = \begin{bmatrix} R_0 & R_1 & R_2 \\ 0 & 0 & 0 \end{bmatrix}, \quad \bar{\Gamma}_{15} = \begin{bmatrix} P_1 \bar{B} \\ W \bar{B} \end{bmatrix}, \quad \bar{\Gamma}_{16} = \begin{bmatrix} -\bar{B}_f \\ -\bar{B}_f \end{bmatrix}$$

$$\bar{\Gamma}_{22} = \text{diag} \{M_0 - Q_0, M_1 - Q_1, M_2 - Q_2\}$$

$$- \text{diag} \{R_0 + N_0, R_1 + N_1, R_2 + N_2\}$$

$$\bar{\Gamma}_{23} = \text{diag} \{N_0 - G_0, N_1 - G_1, N_2 - G_2\}$$

$$\bar{\Gamma}_{24} = \text{diag} \{G_0, G_1, G_2\}$$

$$\bar{\Gamma}_{33} = \text{diag} \{[G_0 - N_0]_s, [G_1 - N_1]_s, [G_2 - N_2]_s\}$$

$$- \text{diag} \left\{ \frac{\pi^2}{4} S_0, \frac{\pi^2}{4} S_1, \frac{\pi^2}{4} S_2 \right\} + \varepsilon \bar{C}_\alpha^T \Omega \bar{C}_\alpha$$

$$\bar{\Gamma}_{34} = \text{diag} \{N_0 - G_0, N_1 - G_1, N_2 - G_2\}$$

$$\bar{\Gamma}_{44} = -\text{diag} \{M_0 + N_0, M_1 + N_1, M_2 + N_2\}$$

$$\bar{\Lambda} = [\bar{A}_1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ \bar{B} \ 0]$$

$$\bar{\Gamma}_j^R = (\tau_m + jT/2) \bar{A}^T R_j, \quad \bar{\Gamma}_j^N = d \bar{A}^T N_j$$

$$\bar{\Gamma}_j^S = (\tau_M + jT/2) \bar{A}^T S_j, \quad \bar{A} = [0 \ \bar{B}_f C]$$

$$\bar{E}_2 = -\bar{D}_f, \quad \bar{C}_\alpha = [\alpha_1 \bar{C}_0 \ \alpha_2 \bar{C}_0 \ \alpha_3 \bar{C}_0]$$

$$\bar{C}_0 = [0 \ C], \quad \bar{C} = [0 \ \bar{D}_f C].$$

Moreover, if the inequality (28) is feasible, the FDF parameters can be obtained by

$$\begin{bmatrix} A_f & B_f \\ C_f & D_f \end{bmatrix} = \begin{bmatrix} W^{-1} & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \bar{A}_f & \bar{B}_f \\ \bar{C}_f & \bar{D}_f \end{bmatrix} \quad (29)$$

or

$$\begin{bmatrix} A_f & B_f \\ C_f & D_f \end{bmatrix} = \begin{bmatrix} \bar{A}_f & \bar{B}_f \\ \bar{C}_f & \bar{D}_f \end{bmatrix} \begin{bmatrix} W^{-1} & 0 \\ 0 & I \end{bmatrix}. \quad (30)$$

Proof: Define matrix $P = \begin{bmatrix} P_1 & P_2 \\ * & P_3 \end{bmatrix}$, and matrix $F = \text{diag} \{I, P_2 P_3^{-1}, I, I, I, I, I, I, I, I, I, I, I, I, I\}$. Then, we perform a congruence transformation to inequality (22), by the matrix F on the left and the matrix F^T on the right.

Define new matrix variables $W = P_2 P_3^{-1} P_2^T$, $\bar{A}_f = P_2 A_f P_3^{-1} P_2^T$, $\bar{B}_f = P_2 B_f$, $\bar{C}_f = C_f P_3^{-1} P_2^T$, and $\bar{D}_f = D_f$. It is obvious that inequality (28) follows the equivalence to inequality (22). In addition, $P > 0$ is equivalent to $P_1 - P_2 P_3^{-1} P_2^T > 0$, i.e., $P_1 - W > 0$. Following the process of proof of Theorem 1, we can derive that the H_∞ performance of the residual system could be guaranteed by (28).

Moreover, since $W = P_2 P_3^{-1} P_2^T > 0$, P_2 is nonsingular. The transfer function from $\bar{y}_m(t)$ to $r(t)$ can be represented by

$$\begin{aligned} T_{\bar{y}_m r} &= C_f(sI - A_f)^{-1} B_f + D_f \\ &= \bar{C}_f P_2^{-T} P_3(sI - P_2^{-1} \bar{A}_f P_2^{-T} P_3)^{-1} P_2^{-1} \bar{B}_f + \bar{D}_f \\ &= \bar{C}_f(sW - \bar{A}_f)^{-1} \bar{B}_f + \bar{D}_f \\ &= \bar{C}_f(sI - W^{-1} \bar{A}_f)^{-1} \bar{B}_f + \bar{D}_f \\ &= \bar{C}_f W^{-1}(sI - \bar{A}_f W^{-1})^{-1} \bar{B}_f + \bar{D}_f. \end{aligned}$$

Then, the FDF parameter matrices can be obtained by (29) or (30). ■

Remark 3: By simply setting $\alpha_1 = 1$, $\alpha_2 = 0$, $\alpha_3 = 0$, and $T = 0$, the condition we developed in Theorem 2 would degrade into FDF design criterion under PETC (8). In other words, the FDF design with PETC is a special case of integral-based event-triggered FDF design method proposed in this paper. Compared with PETC, the PIB-ETC could additionally reduce the amount of data sent. A detailed comparison will be provided in following section.

IV. SIMULATION

In this section, the practical parameters of a USV are borrowed from [15] to demonstrate the effectiveness of the proposed method, and they are listed as follows:

$$\begin{aligned} T_v &= 0.5263, & T_r &= 0.4211, & K_{dr} &= -0.0103 \\ K_{dp} &= -0.0202, & K_{dv} &= 0.0380, & K_{vp} &= 0.7980 \\ K_{vr} &= -0.4600, & \omega_n &= 1.6300, & \zeta &= 2.0840. \end{aligned}$$

We assume that the matrices B_2 and C in (3) are given by $B_2 = [0.6 \ -1 \ 2 \ 0.8 \ 1]^T$ and $C = [1 \ 0.8 \ 1 \ -1 \ 0.6]$, respectively. The state feedback controller is predesigned as $K = [1.1118 \ 3.4650 \ 6.6327 \ 1.5691 \ 3.1662]$. The network-induced delay satisfies $\eta_m \leq \eta_k \leq \eta_M$ with $\eta_m = 10$ ms and $\eta_M = 50$ ms.

The parameters of fault weighting system are chosen as follows:

$$A_w = -0.1, \quad B_w = 0.25, \quad C_w = 0.5, \quad D_w = 0. \quad (31)$$

Suppose that $T = 0.4$ s and $T = 0.6$ s. For different ETC parameters pair (h, ε) , solve the following optimization problem:

$$\begin{aligned} \min \quad & \gamma \\ \text{s.t.} \quad & \text{conditions in (28)}. \end{aligned} \quad (32)$$

The obtained optimal H_∞ performance indices γ^* are listed in Table I for $T = 0.4$ s and Table II for $T = 0.6$ s, respectively. It is clear that a larger h or T will result in a larger γ^* , and a larger ε will result in a larger γ^* as well, which is reasonable since a larger h , T , or ε make the ETC in (7) more difficult to be triggered, i.e., less system information can be received by FDF.

For the purpose of evaluating the fault detection ability of the proposed method, we assume that the external disturbance $\omega_\psi(t)$ is $\mathcal{N}(0, 1)$ (standard Gaussian distributed random signal) and the disturbance $\omega_\phi(t)$ is assumed as $\omega_\phi(t) = 0.5 \sin(0.3t)$.

TABLE I
 γ^* UNDER DIFFERENT ETC WITH $T = 0.4$

ETC	h	0.4			0.6		
	ε	0.2	0.3	0.4	0.2	0.3	0.4
γ^*		0.8274	0.8924	0.9511	0.8325	0.8951	0.9546

TABLE II
 γ^* UNDER DIFFERENT ETC WITH $T = 0.6$

ETC	h	0.4			0.6		
	ε	0.2	0.3	0.4	0.2	0.3	0.4
γ^*		0.8274	0.8924	0.9511	0.8325	0.8951	0.9546

TABLE III
DATA TRANSMISSION UNDER DIFFERENT SAMPLING SCHEME

	h	ε	T	SM	$R_{e/t}$
Periodic Sampling	0.4	—	—	50.0	100%
PETC	0.4	0.3	—	18.4	36.8%
Theorem 2	0.4	0.3	0.4	9.9	19.8%
Theorem 2	0.4	0.3	0.6	9.7	19.4%

We suppose that the system output $y(t)$ is contaminated by random measurement noise that is expressed by $\mathcal{N}(0, 0.5)$, and the fault signal $f(t)$ is described as

$$f(t) = \begin{cases} 1, & t \in [6, 12] \\ 0, & \text{otherwise} \end{cases}. \quad (33)$$

Choose ETC parameters as $h = 0.4$ and $\varepsilon = 0.3$. According to Theorem 2, the event-triggered matrix Ω and the parameters of the proposed FDF can be obtained.

To have a fair evaluation of the system, numerical experiment is performed for $t \in [0, 20]$ under different initial conditions given by

$$x(0) = \begin{bmatrix} \cos(2\pi k/30) \\ -\cos(2\pi k/30) \\ \sin(2\pi k/30) \\ -\sin(2\pi k/30) \\ 0 \end{bmatrix}, \quad k = 1, 2, \dots, 30. \quad (34)$$

The average amounts of data transmission under different sampling scheme and different ETC parameters are listed in Table III, where SM is the average amounts of sent measurements and $R_{e/t}$ denotes the ratio between the quantity of data transmission in the event-triggering scheme and the corresponding quantity under periodic sampling. It is obvious that compared with the traditional event-triggering scheme (8) and periodic sampling, the quantity of data transmission by PIB-ETC (7) has been reduced significantly. Specifically, compared with periodic sampling and periodic event trigger, the utilization of bandwidth by using PIB-ETC with $T = 0.4$ is reduced 80.2% and 17.0%, respectively. With same the h and ε , a larger T will lead to a

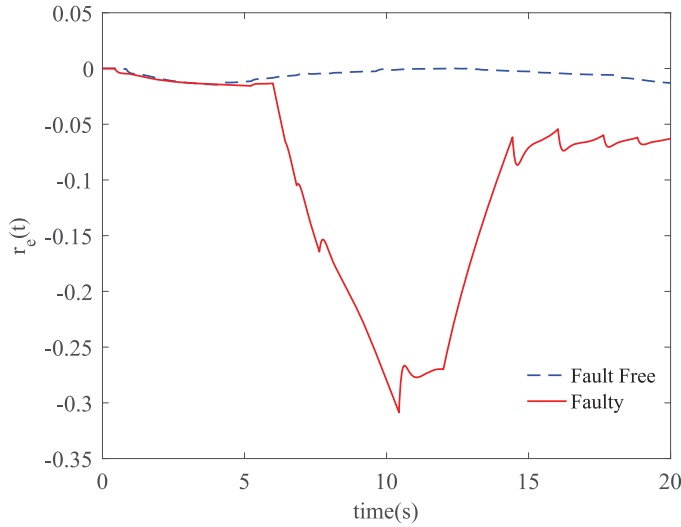


Fig. 3. Residual response $r_e(t)$ with $T = 0.4$, $h = 0.4$, and $\varepsilon = 0.3$.

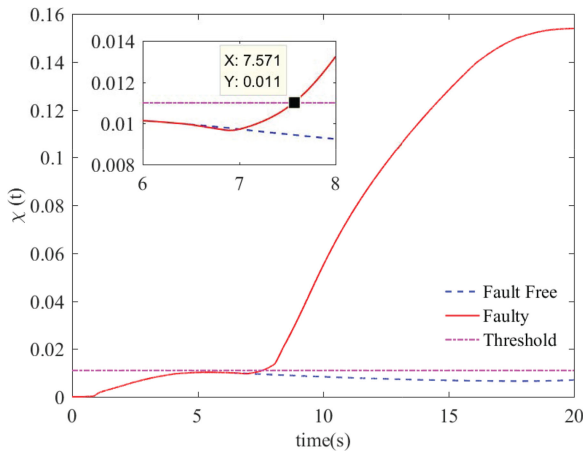


Fig. 4. Residual evaluation response $\chi(t)$ $T = 0.4$, $h = 0.4$, and $\varepsilon = 0.3$.

lower bandwidth utilization since it makes PIB-ETC in (7) more difficult to be triggered.

Set $k = 1$ in (34). The residual response signal $r_e(t)$ and the residual evaluation function response $\chi(t)$ are depicted in Figs. 3 and 4, respectively. The threshold for $\chi(t)$ is chosen as $\chi_{th} = 0.011$. It is observed that system fault in (33) is detected at $t = 7.571$ s, i.e., the fault detection time in Fig. 4 is 1.571 s. An alarm of fault will be generated after the occurrence of fault is detected. Fig. 5 displays the release instants and the release intervals of system, from which one can see that only 11 measurements are transmitted through the communication network. Thus, the bandwidth occupation of network communication can be reduced significantly. In summary, the proposed method can not only detect the occurrence of fault in time, but also save the communication resources effectively.

To illustrate the relationship between the amount of sent measurements and fault detection performance, choose PIB-ETC parameters as $T = 0.4$, $h = 0.2$, and $\varepsilon = 0.1$. By Theorem 2, the matrix Ω and the parameters of FDF can be obtained. Under PIB-ETC parameters $T = 0.4$, $h = 0.2$, and $\varepsilon = 0.1$, the

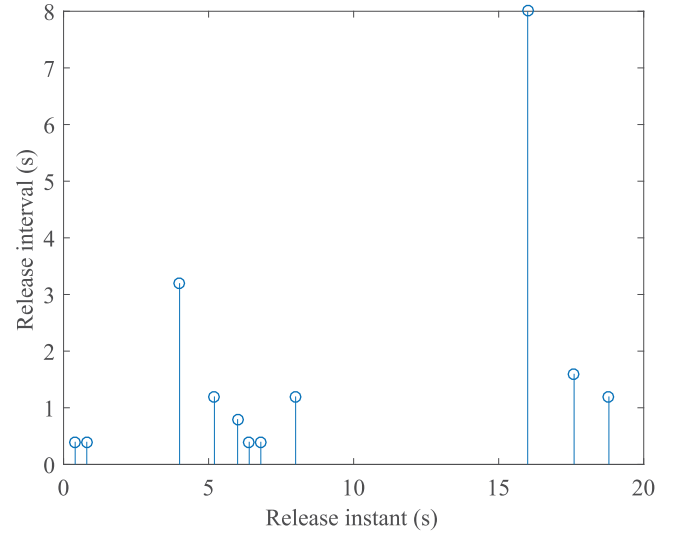


Fig. 5. Evolution of event trigger $T = 0.4$, $h = 0.4$, and $\varepsilon = 0.3$.

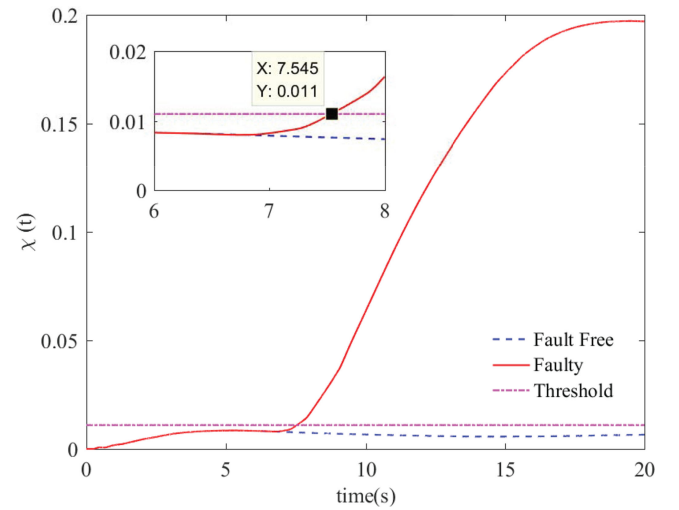


Fig. 6. Residual evaluation response $\chi(t)$ with $T = 0.4$, $h = 0.2$, and $\varepsilon = 0.1$.

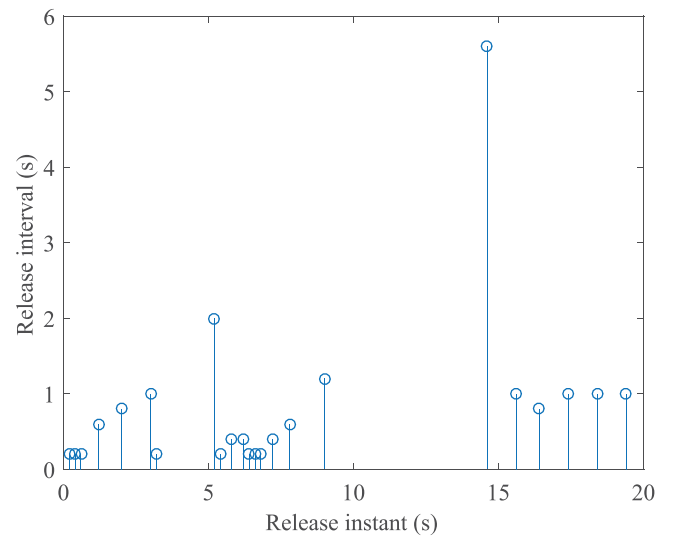


Fig. 7. Evolution of event trigger with $T = 0.4$, $h = 0.2$, and $\varepsilon = 0.1$.

residual evaluation function response $\chi(t)$ with fault signal $f(t)$ in (33) is depicted in Fig. 6. The threshold for $\chi(t)$ is chosen as $\chi_{th} = 0.011$ as well. It can be observed that the system fault is detected at $t = 7.545$ s, i.e., the fault detection time in Fig. 6 is 1.545 s, which is smaller than the fault detection time under PIB-ETC with $T = 0.4$, $h = 0.4$, and $\varepsilon = 0.3$. The release instants and the release intervals of system under PIB-ETC parameters $T = 0.4$, $h = 0.2$, and $\varepsilon = 0.1$ is shown in Fig. 7, from which one can see that 23 measurements are transmitted. Thus, we can conclude that there is a tradeoff between amounts of sent measurement and fault detection time. Specifically, a smaller h and a smaller ε could lead to a smaller fault detection time.

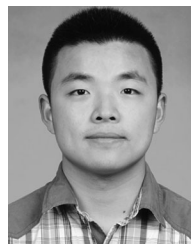
V. CONCLUSION

In this paper, a framework of event-based fault detection for a USV is established. The integral-based ETC and FDF coordinated design method is provided. Compared with existing results, the proposed method can not only detect the system fault in time, but also reduce the occupation of network bandwidth more effectively. The effectiveness of the proposed approach is demonstrated by simulation results.

On the basis of the research done in this paper, there are many interesting and challenging topics deserving to be studied in the future. For example, if the packet dropout is considered, then the integral-based ETC should be redesigned by considering the occurrence of packet dropout to make the system more robust. Moreover, the co-design of controller, FDF, and integral-based ETC is also a challenging and meaningful work, which may further help reduce the conservatism of the proposed method.

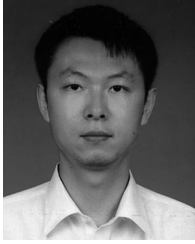
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